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BSc Econometrics and Data Science

The Time-Varying Effect of Monetary Policy on Inflation: Evidence from ECB and Federal Reserve Leadership Regimes

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June 23, 2026

Statement of Originality

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Abstract

This thesis examines whether the effect of monetary policy on inflation varies across central bank leadership regimes and identifies structural breaks in the relationship between interest rates and inflation using the Bai–Perron methodology. Using monthly data from the European Central Bank and the Federal Reserve, the study investigates how changes in policy interest rates affect inflation and whether this relationship changes over time. A leadership regime model is estimated using Two Stage Least Squares regression with interaction terms between lagged interest rates and leadership dummies. HAC standard errors are applied to account for heteroskedasticity and autocorrelation. Additionally, a Bai–Perron structural break analysis is conducted to identify changes in the relationship between interest rates and inflation directly from the data and to compare them with leadership transitions. The results indicate that the effect of monetary policy on inflation is not constant over time. For the Federal Reserve in leadership regime model, several leadership interactions are statistically significant and the joint Wald test provides strong evidence of time-varying effects. For the ECB, the evidence is weaker, with only limited support for differences across leadership periods. The structural break analysis identifies multiple significant breaks for both central banks, many of which correspond more closely to major macroeconomic events than to leadership changes. Overall, the findings suggest that monetary policy effects on inflation are time-varying, although leadership shifts alone cannot fully explain the observed variation.

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1 Introduction

In modern macroeconomics, central banks mainly implement monetary policy by adjusting interest rates, which are now the most widely used policy instrument. A previous study by Vegh (2001) points out that short-term nominal interest rates are increasingly regarded as the main instrument of monetary policy by policymakers, often together with some inflation target. Interest rates are not only a policy tool, but also a way to keep inflation under control. More specifically, interest rate changes influence the economy through the monetary transmission process, meaning that monetary policy decisions lead to changes in real GDP and inflation (McCallum, 1999). Evidence from the euro area suggests that an unexpected and temporary increase in the short-term interest rate leads to a temporary decrease in output after a few quarters, while the price level takes longer to fall (Peersman & Smets, 2003). Overall, monetary policy can affect inflation through changes in interest rates and their influence on economic activity, even if the effect may be delayed.

However, the relationship between monetary policy and inflation may not be constant over time. Monetary policy does not remain fixed, and policymakers' preferences may shift across different periods. As a result, the effect of monetary policy on inflation can vary over time. Trecroci and Vassalli (2010) suggest that the monetary policy rules should not be assumed to be constant over time, as a constant parameter reaction function may ignore real-world factors such as conflicting objectives and policymakers' preferences. Similarly, Davig and Doh (2014) argue that monetary policy may shift across different regimes, with shifts in the monetary policy reaction coefficients. They also find that more aggressive adjustments of the nominal interest rate in response to inflation can reduce inflation persistence. Taken together, the impact of interest rates on inflation may not be the same over time, and may depend on how monetary policy is conducted. Because monetary policy can vary across different periods, the effect may also be different across different central bank leadership regimes.

While existing studies have examined the relationship between monetary policy and inflation using interest rate changes as a proxy for monetary policy, relatively few studies investigate how this relationship may vary across different central bank leadership periods. Based on these previous studies, my research question is: How does the effect of mone-

tary policy on inflation vary across central bank leadership regimes? This study aims to explore whether the impact of monetary policy on inflation varies across different leadership periods using interest rate changes as a monetary policy instrument, and to compare these differences using regression analysis based on data from the euro area and the United States.

In this paper, each central bank president is treated as a separate monetary policy period. Different leaders often respond to economic problems in different ways. For example, Paul Volcker raised interest rates aggressively to reduce high inflation in the US from 1979 to 1987. In contrast, Ben Bernanke, who took office from 2006 to 2014, kept interest rates very low and introduced quantitative easing after the 2008 financial crisis to support the recovery of the economy. These differences suggest that shifts in central bank leadership can lead to changes in monetary policy and affect how monetary policy influences inflation.

In the following chapters, this paper first reviews the relevant literature, followed by a description of the data and methodology. The results are then presented and discussed, and the final chapter concludes.

2 Literature Review

The relationship between interest rates and inflation has been a crucial topic in macroeconomics for a long time. Fisher (1930) proposed that nominal interest rates reflect real interest rates and expected inflation. A large number of empirical studies have proved this relationship and generally find that interest rates and inflation are closely related (Mishkin, 1992; Mishkin & Simon, 1995; Westerlund, 2008).

While a large body of literature supports the existence of a relationship between interest rates and inflation, a number of studies focus on how this relationship works. Mishkin (1996) gives a detailed description of the monetary transmission mechanism through different channels, and shows that through the traditional interest rate channel, an expansionary monetary policy decreases the interest rate, then brings an increase in investment spending and finally leads to a rise in aggregate demand and output. However, inflation is not determined by the level of output, but by the output gap, which is the difference between the actual output and potential output. When actual output changes relative to potential

output, the output gap changes. In this case, inflation tends to change since inflation is positively related to the output gap, which is suggested by the New Keynesian Phillips Curve (Galí, 2016). This implies that interest rate changes, as an important instrument of monetary policy, have an impact on inflation through their effect on the output gap.

However, this mechanism does not always work as expected. Increases in interest rates by former Federal Reserve Chairman Martin failed to reduce inflation, as government borrowing to finance the Vietnam War and the Great Society programs continued to support aggregate demand through a fiscal transmission channel. A similar situation occurred during the COVID-19 period. Even if the Federal Reserve had maintained somewhat higher interest rates during the pandemic stimulus, inflation might still have emerged if it had continued to purchase government debt in order to sustain those interest rates (Gmeiner & Larson, 2023). These findings suggest that the relationship between monetary policy and inflation depends not only on monetary transmission mechanisms but also on fiscal policy, raising the question of whether the impact of monetary policy on inflation remains stable over time. This concern is also reflected in the empirical literature. Cogley and Sargent (2001) show that changes in the law of motion for inflation and unemployment are related to shifts in monetary authority behaviour, which in turn arise from changes in policymakers' beliefs. Their findings suggest that the relationship between monetary policy and inflation may not remain stable over time.

One possible reason for this instability may be that monetary policy has not been implemented in the same way over time, but has changed across different central bank leadership periods. Clarida et al. (2000) illustrate that monetary policy was implemented in significantly different ways before and after Paul Volcker was appointed Chairman of the Board of Governors of the Federal Reserve System. In the earlier period, nominal interest rates increased by less than expected inflation, whereas in the Volcker-Greenspan period they increased by more than expected inflation. Lubik and Schorfheide (2004) study a similar case and find that "U.S. monetary policy post-1982 is consistent with determinacy, whereas the pre-Volcker policy is not" (p.190). These studies suggest that monetary policy varies across different leadership periods and alter its impact on inflation.

In addition to monetary policy, the relationship between unemployment rate and in-

flation are also widely discussed. Phillips (1958) finds a negative relationship between unemployment and wage inflation. This relationship is later improved by Friedman (1968), who suggests that the trade-off between unemployment and inflation only exists in the short run.

Overall, the literature shows that monetary policy is related to inflation, but its effect on inflation is not always consistent. While many studies focus on the monetary transmission mechanism or how monetary policy changes across periods, they usually do not test whether the effect of monetary policy on inflation differs across central bank leadership regimes. Therefore, this thesis investigates whether this effect is time-varying across different leadership periods, using interest rates as a measure of monetary policy, based on data from the European Central Bank (ECB) and the Federal Reserve (Fed).

3 Methodology

3.1 Data Description

This study uses data from both the euro area and the United States to test the effect of monetary policy on inflation.

For the euro area, data on the main refinancing operations rate, the Harmonized Index of Consumer Prices (HICP) inflation rate, unemployment rate and industrial production index are obtained from ECB Data Portal. The policy interest rate is presented on a daily basis and is converted to monthly frequency by taking the monthly average to match the frequency of inflation data, similar to the treatment of interest data in Bernanke and Blinder (1992). The data start in April 2000 and end in February 2026.

For the United States, data on the federal funds effective rates, consumer price index (CPI), unemployment rate and industrial production index are obtained from Federal Reserve Economic Data. Since the inflation data are not directly available, it can be calculated by taking the log difference of CPI. The data start in February 1970 and end in February 2026.

The sample periods differ because the available data for the United States begin earlier than those for the euro area. The longer U.S. sample is useful for testing time-varying

changes in the influence of monetary policy on inflation over a longer period, while the post-2000 period can be used for comparison with the euro area. All variables are used at a monthly frequency and missing values are filled using linear interpolation.

3.2 Baseline Model

To examine the relationship between monetary policy and inflation, the following baseline model is considered:

$$\pi_t = \alpha + \beta i_{t-1} + \gamma u_t + \epsilon_t \tag{1}$$

where π_t represents inflation, i_{t-1} is the lagged policy interest rate and u_t is the unemployment rate. The error term ϵ_t is assumed to have zero mean and constant variance. The lagged interest rate is used because changes in monetary policy usually affect inflation with a delay rather than within the same month. Batini and Nelson (2001) find that monetary policy actions may take more than one year to fully affect inflation, which supports the use of a lagged interest rate in the model.

This baseline model provides the foundation for the subsequent analysis. The following subsections extend this model by allowing the effect of interest rates on inflation to vary across leadership regimes and structural break periods.

3.3 Leadership Regime Model

To examine whether the effect of monetary policy on inflation differs across central bank leadership periods, the baseline model is extended to the form (Wooldridge, 2021):

$$\pi_t = \alpha + \beta i_{t-1} + \sum_{j=1}^{J} \delta_j (i_{t-1} \times D_j) + \gamma u_t + \epsilon_t \tag{2}$$

where D_j is a dummy variable for each leadership period. Each D_j equals 1 when the jth central bank president is in office and 0 otherwise. The interaction terms allow the effect of interest rates on inflation to vary across different monetary policy regimes.

The lagged interest rate is potentially endogenous because it may correlate with the disturbance. Since i_{t-1} was determined in period $t-1$, it may still be affected by the

previous error term. If the disturbance term is persistent and follows an AR process,

$$\epsilon_t = \rho\epsilon_{t-1} + \eta_t \tag{3}$$

where $|\rho| < 1$ and η_t is a white noise (Wooldridge, 2021). Then the current error term ϵ_t is partly related to the previous error ϵ_{t-1}. As a result, i_{t-1} may be correlated with ϵ_t. Additionally, unemployment may also be endogenous according to the Phillips Curve (Galí & Gertler, 1999), inflation and unemployment are jointly determined. To deal with this issue, unemployment is treated as another endogenous regressor.

Therefore, the model is estimated using TSLS regression. In this paper, the instrument set is defined as,

$$Z_t = (1, \pi_{t-2}, \pi_{t-3}, u_{t-2}, u_{t-3}, i_{t-2}, i_{t-3}) \tag{4}$$

The instrument set is likely to be relevant because past values of inflation, unemployment and interest rates are closely related to current monetary policy decisions and labour market conditions. As a result, these variables are expected to be correlated with the endogenous variables i_{t-1} and u_t. It is also assumed to satisfy the exogeneity condition. Since all instruments are lagged by at least two periods, they are predetermined with respect to the current disturbance term ϵ_t. Under the assumption that shocks do not persist for many periods, these lagged variables should not be correlated with the current error term. Therefore, they can be used as valid instruments in the TSLS estimation.

In the first stage, the endogenous regressors are projected onto the instrument set,

$$i_{t-1} = Z_t'\delta_1 + v_{1t} \tag{5}$$

$$u_t = Z_t'\delta_2 + v_{2t} \tag{6}$$

which gives the fitted values $\hat{i}_{t-1}$ and $\hat{u}_t$. In the second stage, these fitted values replace the original endogenous variables in the leadership regime model, and after accounting for endogeneity, the model estimated in the second stage becomes:

$$\pi_t = \alpha + \beta\hat{i}_{t-1} + \sum_{j=1}^{J} \delta_j(\hat{i}_{t-1} \times D_j) + \gamma\hat{u}_t + \epsilon_t \tag{7}$$

Since the analysis uses monthly time-series data, HAC standard errors are used to account for possible autocorrelation and heteroskedasticity in the error term. Wald tests are then used to examine whether the interaction terms are jointly significant across different leadership periods. The euro area and the United States are analyzed separately.

3.4 Structural Break Analysis

To compare the leadership-based regimes with break points identified from the data, a structural break analysis is also conducted using the baseline model. Rather than setting the break dates according to central bank leadership periods, the break points are estimated directly from the data. Following Boldea and Hall (2025), the timing and number of break points are treated as unknown. The estimated break dates are then compared with the tenure periods of central bank presidents to examine whether major policy changes identified from the data occur around leadership transitions.

The fitted values obtained from the first-stage regressions described in the previous subsection are used in the structural break analysis. Following Hall et al. (2012), structural breaks are estimated within the TSLS framework in order to account for the potential endogeneity of interest rates and unemployment.

First, structural breaks in the first-stage equations are estimated using the Bai-Perron method (Bai & Perron, 1998). The Bai-Perron procedure identifies optimal break dates by selecting the break specification that minimizes the residual sum of squares (RSS). The preferred number of breaks is then chosen using the Bayesian Information Criterion (BIC). After the break dates have been identified, the first-stage equations are re-estimated within each regime and fitted values $\hat{i}_{t-1}$ and $\hat{u}_t$ are obtained for the endogenous regressors. The fitted values are then used in the second-stage equation. Following Hall et al. (2012), structural breaks in the second-stage equation are also treated as unknown and are estimated using the Bai-Perron procedure. The number of structural breaks is determined by selecting the alternative with the lowest BIC.

After the break dates have been identified, their statistical significance is evaluated using Bai-Perron tests for multiple breaks at known break dates (Bai & Perron, 1998). In addition to the overall multiple-break test, individual tests are conducted for each estimated

break date. The estimated breaks and the corresponding coefficient estimates are reported in the Results section.

3.5 Variable Description

This study uses the following variables.

Table 1: Variable Definitions

Variable	Description	Measurement
π_t	Inflation	HICP inflation rate for the euro area; log difference of CPI for the United States
i_{t-n}	Lagged interest rate	Main refinancing operations rate for the euro area; effective federal funds rate for the United States
u_t	Unemployment	Unemployment rate
D_t	Regime dummy	Dummy variable representing different central bank leadership periods

Regime variables are constructed based on the tenure periods of ECB presidents and Fed Chairs. Each dummy variable equals 1 during the corresponding leadership period and 0 otherwise.

For the euro area, the data is available from April 2000. Data for 1998 and 1999 and some for 2000 during the tenure of the first ECB president, Wim Duisenberg, is unavailable. The ECB leadership regimes are shown in Table 2 and the Federal Reserve leadership regimes are shown in Table 3.

Table 2: ECB Leadership Regimes

Dummy	Period	President
D_1	Apr 2000 – Oct 2003	Wim Duisenberg
D_2	Nov 2003 – Oct 2011	Jean-Claude Trichet
D_3	Nov 2011 – Oct 2019	Mario Draghi
D_4	Nov 2019 – Feb 2026	Christine Lagarde

Table 3: Federal Reserve Leadership Regimes

Dummy	Period	Fed Chair
D_1	Feb 1970 – Jan 1978	Arthur Burns
D_2	Feb 1978 – Aug 1979	William Miller
D_3	Aug 1979 – Aug 1987	Paul Volcker
D_4	Aug 1987 – Jan 2006	Alan Greenspan
D_5	Feb 2006 – Jan 2014	Ben Bernanke
D_6	Feb 2014 – Feb 2018	Janet Yellen
D_7	Feb 2018 – Feb 2026	Jerome Powell

4 Results

The first part of this section reports the results of the leadership regime model estimated for the ECB and the Federal Reserve. Interaction terms between the lagged interest rate and leadership dummies are used to detect potential changes in the effect of monetary policy across different leadership periods. HAC standard errors are used to account for possible autocorrelation and heteroskedasticity, while joint F-tests are conducted to examine whether the interaction effects are jointly significant. The second part presents the results of the structural break analysis. Following Hall et al. (2012), structural breaks in both the first-stage and second-stage equations are estimated using the Bai-Perron methodology (Bai & Perron, 2003). The estimated break dates are then evaluated using Bai-Perron tests for multiple breaks at known break dates (Bai & Perron, 1998). The final part presents the result for instrument diagnostic tests proving if the instruments are relevant, strong and

exogenous.

4.1 Result for Leadership Regime Model

Tables 4 and 5 report the TSLS estimates with HAC standard errors for the ECB and the Federal Reserve. For the ECB, the coefficient of the lagged interest rate is positive (0.1182) but statistically insignificant, suggesting that there is no clear evidence of a systematic relationship between interest rates and inflation during the baseline leadership period. Among the interaction terms, only the Draghi coefficient is positive and statistically significant at the 5% level (2.5640), indicating that the estimated effect of interest rates on inflation was stronger during the Draghi period than during the Duisenberg period. The coefficients for Trichet and Lagarde are not statistically significant. The unemployment coefficient is negative but statistically insignificant.

For the Federal Reserve, the coefficient of the lagged interest rate is positive (0.6981) and statistically significant at the 1% level, indicating a positive relationship between interest rates and inflation during the baseline period. Among the interaction terms, the coefficient for Greenspan is negative and statistically significant at the 5% level, while the coefficients for Bernanke and Powell are significant at the 10% level. These negative coefficients suggest that the effect of interest rates on inflation was weaker during these leadership periods relative to the Burns period. The coefficients for Miller, Volcker and Yellen are not statistically significant under HAC standard errors. Similar to the ECB results, the unemployment coefficient is negative but statistically insignificant.

Table 4: ECB Results (HAC)

Variable	Coefficient
Intercept	8.4234
	(5.6677)
Lagged interest rate	0.1182
	(0.2466)
Trichet interaction	-0.0504
	(0.2384)
Draghi interaction	2.5640**
	(1.2984)
Lagarde interaction	-0.1915
	(0.2535)
Unemployment	-0.7373
	(0.5408)
Observations	311
Adjusted R^2	0.2850

Table 5: Fed Results (HAC)

Variable	Coefficient
Intercept	2.8135
	(3.1987)
Lagged interest rate	0.6981***
	(0.1872)
Miller interaction	0.0674
	(0.1453)
Volcker interaction	-0.2392
	(0.2415)
Greenspan interaction	-0.4528**
	(0.1931)
Bernanke interaction	-0.4592*
	(0.2704)
Yellen interaction	-1.1565
	(1.8436)
Powell interaction	-0.4417*
	(0.2547)
Unemployment	-0.1634
	(0.3853)
Observations	673
Adjusted R^2	0.6101

Notes: TSLS estimates with HAC standard errors computed using the Quadratic Spectral kernel. Duisenberg is the reference period for the ECB and Burns is the reference period for the Federal Reserve. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The results of the joint tests for leadership interaction effects are reported in Table 6. For the ECB, the null hypothesis that all interaction coefficients are jointly equal to zero cannot be rejected ($\chi^2 = 5.5213$, $p = 0.1374$). This suggests that the interaction effects are

not jointly significant after accounting for endogeneity and HAC standard errors. Although the Draghi interaction term is individually significant in the previous regression results, the joint test indicates that the interaction terms do not provide sufficient evidence of systematic differences across ECB leadership periods as a whole. For the Federal Reserve, the null hypothesis is strongly rejected ($\chi^2 = 53.141$, $p = 1.099 \times 10^{-9}$), indicating that the effect of interest rates on inflation varies across leadership periods. The result is consistent with the individual coefficient estimates, where several leadership interaction terms are statistically significant. Overall, the evidence for time-varying monetary policy effects appears stronger for the Federal Reserve than for the ECB.

Table 6: Joint Tests for Leadership Interaction Effects

Model	Tested coefficients	χ^2 statistic	p-value
ECB	Trichet, Draghi, Lagarde	5.5213	0.1374
Federal Reserve	Miller, Volcker, Greenspan, Bernanke, Yellen, Powell	53.1410	1.099e-09

Notes: The null hypothesis is that all leadership interaction terms are jointly equal to zero, i.e. $H_0 : \delta_1 = \delta_2 = \cdots = \delta_n = 0$. The tests are Wald tests based on TSLS estimates with HAC standard errors computed using the Quadratic Spectral kernel.

As a robustness check, the ECB model is re-estimated without the leadership interaction terms. The restricted model produces similar results: the coefficient of the lagged interest rate remains positive but statistically insignificant (0.0791, $p = 0.8388$), while the unemployment coefficient remains negative and statistically insignificant (-0.4761, $p = 0.3870$). Compared with the baseline specification, the signs of both coefficients remain unchanged and their statistical significance does not differ. These results suggest that the ECB findings are not sensitive to the inclusion of the leadership interaction terms. Together with the largely insignificant interaction coefficients, these provide further support for the conclusion that there is limited evidence of differences in the effect of monetary policy across ECB leadership periods.

4.2 Result for Structural Break Analysis

Table 7: Pseudo-code for Structural Break Selection

Algorithm: Bai-Perron Structural Break Selection

1. Estimate the first-stage equations for the endogenous variables i_{t-1} and u_t using the selected instruments.
2. For each equation, apply the Bai-Perron procedure and allow the number of breaks to vary from 0 to the maximum number of breaks.
3. For each possible number of breaks, record the corresponding BIC value.
4. Select the number of breaks that gives the lowest BIC value.
5. Store the estimated break dates from the preferred specification.
6. Re-estimate the first-stage equations within the identified regimes and obtain the fitted values $\hat{i}_{t-1}$ and $\hat{u}_t$.
7. Use the fitted values in the second-stage equation.
8. Apply the Bai-Perron procedure again to the second-stage equation.
9. Select the preferred number of second-stage breaks using the lowest BIC value.
10. Report the selected break dates and test their statistical significance.

Before reporting the estimated break dates, the procedure used to select the number and timing of structural breaks is summarized in Table 7. Several models for first stage and second stage with different numbers of breaks were estimated and compared. The BIC values were then used to evaluate which alternative provided the best fit to the data. In some cases, adding further breaks changed the BIC only slightly. The final specification was selected based on the lowest BIC and was used to obtain the break dates discussed in the following contents.

The structural break dates identified by the Bai-Perron procedure for the first-stage equations are reported in Tables 8 and 9. For the ECB lagged interest rate equation, three structural breaks are selected by the lowest BIC, occurring in December 2002, October

2008 and August 2022. In the ECB unemployment equation, six breaks are identified in October 2005, August 2008, May 2011, December 2013, March 2020 and October 2022. For the Federal Reserve, two breaks are selected in the interest rate equation, occurring in October 1980 and June 1987, while one break is identified in the unemployment equation in March 2020. Obviously, more breaks are identified for the ECB than for the Federal Reserve. The contrast is strongest in the unemployment equation, where six breaks are found for the ECB but only one for the Federal Reserve. The ECB results therefore indicate more frequent changes over time in the first-stage relationships.

Table 8: Structural Breaks in the First-Stage Interest Rate Equation

Model	Selected Breaks	Break Dates
ECB	3	Dec 2002, Oct 2008, Aug 2022
Federal Reserve	2	Oct 1980, Jun 1987

Notes: Break dates are estimated using the Bai-Perron procedure. The number of breaks is determined using lowest BIC. Detailed BIC values are reported in Appendix.

Table 9: Structural Breaks in the First-Stage Unemployment Equation

Model	Selected Breaks	Break Dates
ECB	6	Oct 2005, Aug 2008, May 2011, Dec 2013, Mar 2020, Oct 2022
Federal Reserve	1	Mar 2020

Notes: Break dates are estimated using the Bai-Perron procedure. The number of breaks is determined using lowest BIC. Detailed BIC values are reported in Appendix.

Table 10 presents the results of the second-stage equation. Based on the lowest BIC, five breaks are found for both ECB and the Federal Reserve. For the ECB, the estimated break dates are July 2007, February 2010, November 2016, May 2020 and December 2022. For the Federal Reserve, the estimated break dates are November 1980, February 1987, December 1999, October 2009 and June 2020. The results suggest that the relationship

between interest rates and inflation is not stable throughout the sample period. Instead, the estimated coefficients appear to change across a number of distinct periods. Although both central banks are associated with five estimated breaks, the timing of these breaks differs considerably, indicating that the sources of structural change may not be the same across the two monetary authorities

Table 10: Bai-Perron Structural Breaks in the Second-Stage Equation

Model	Selected Breaks	Break Dates
ECB	5	Jul 2007, Feb 2010, Nov 2016, May 2020, Dec 2022
Federal Reserve	5	Nov 1980, Feb 1987, Dec 1999, Oct 2009, Jun 2020

Notes: Break dates are estimated using the Bai-Perron procedure. The number of breaks is determined using lowest BIC. Detailed BIC values are reported in Appendix.

Table 11 presents the Bai-Perron tests for structural break significance. The null hypothesis of no structural breaks is rejected for all six specifications given the extremely small p-values. In the ECB models, the interest rate first-stage equation contains three breaks ($F = 5.23$, $p < 0.001$), the unemployment first-stage equation contains six breaks ($F = 8.10$, $p < 0.001$), and the second-stage equation contains five breaks ($F = 20.34$, $p < 0.001$). For the Federal Reserve, two breaks are found in the interest rate first-stage equation ($F = 15.47$, $p < 0.001$), one break is identified in the unemployment first-stage equation ($F = 27.98$, $p < 0.001$), and five breaks are found in the second-stage equation ($F = 82.56$, $p < 0.001$). The results confirm that the identified break dates are statistically significant. This supports analyzing the sample data as a series of different periods rather than assuming a single stable relationship throughout.

Table 11: Bai-Perron Tests for Structural Break Significance

Model	Number of Breaks	F-statistic	p-value
ECB i_{t-1} First Stage	3	5.23	< 0.001
ECB u_t First Stage	6	8.10	< 0.001
ECB Second Stage	5	20.34	< 0.001
Fed i_{t-1} First Stage	2	15.47	< 0.001
Fed u_t First Stage	1	27.98	< 0.001
Fed Second Stage	5	82.56	< 0.001

Notes: The table reports Bai-Perron tests for multiple breaks at known break dates. The null hypothesis is that no structural breaks are present. The alternative hypothesis is that the given breaks are present.

Table 12 reports the results of Bai-Perron tests for each break date separately. Using the 5% significance level, the estimated break dates for the ECB show mixed significances. In the interest rate first-stage equation, only the August 2022 break is significant ($F = 6.07$, $p < 0.001$). In the unemployment first-stage equation, the breaks in October 2005, August 2008, December 2013 and October 2022 are significant, while the remaining breaks are not. In the second-stage equation, only the May 2020 break remains statistically significant when tested separately ($F = 12.60$, $p < 0.001$).

For the Federal Reserve, all estimated break dates are statistically significant at the 5% level when tested separately. The largest F-statistic is observed for the November 1980 break in the second-stage equation ($F = 255.75$, $p < 0.001$), while all other breaks also remain significant.

Table 13 reports the estimated interest rate coefficients for each regime identified by the Bai-Perron procedure in the second stage. For the ECB, the coefficient increases from 0.084 before the first break to 1.091 during 2007 to 2010 and 2.146 during 2010 to 2016, reaching its highest value 18.974 during 2016-2020 period before declining and becoming negative -0.827 in the final regime. For the Federal Reserve, the coefficient is positive before 1999, becomes negative during 2000-2009 (-0.185), turns positive again after 2009 (0.627), and becomes negative -0.931 in the final regime. Overall, the estimated coefficients vary considerably across regimes, particularly for the ECB. Compared with the leadership regime model, the structural break analysis shows more evidence of changes in the effect of monetary policy on inflation. This suggests that the observed variation cannot be explained

by leadership changes alone.

Table 12: Individual Bai-Perron Tests for Each Break Date

Model	Break Date	Equation	F-statistic	p-value
ECB	Dec 2002	i_{t-1} first stage	1.48	0.18
ECB	Oct 2008	i_{t-1} first stage	2.01	0.06
ECB	Aug 2022	i_{t-1} first stage	6.07	0.00
ECB	Oct 2005	u_t first stage	2.69	0.01
ECB	Aug 2008	u_t first stage	2.38	0.03
ECB	May 2011	u_t first stage	0.32	0.93
ECB	Dec 2013	u_t first stage	3.98	0.00
ECB	Mar 2020	u_t first stage	1.72	0.12
ECB	Oct 2022	u_t first stage	2.89	0.01
ECB	Jul 2007	second stage	0.76	0.47
ECB	Feb 2010	second stage	2.33	0.10
ECB	Nov 2016	second stage	1.12	0.33
ECB	May 2020	second stage	12.60	0.00
ECB	Dec 2022	second stage	1.22	0.30
Fed	Oct 1980	i_{t-1} first stage	13.59	0.00
Fed	Jun 1987	i_{t-1} first stage	11.93	0.00
Fed	Mar 2020	u_t first stage	27.98	0.00
Fed	Nov 1980	second stage	255.75	0.00
Fed	Feb 1987	second stage	66.60	0.00
Fed	Dec 1999	second stage	14.02	0.00
Fed	Oct 2009	second stage	12.96	0.00
Fed	Jun 2020	second stage	22.21	0.00

Notes: The table reports Bai-Perron tests for each break date separately. Each row tests the null hypothesis of no structural break against the alternative of one break at the specified date. These tests should be interpreted as individual known-break tests rather than as tests of the full multiple-break specification.

Table 13: Second-Stage Coefficient Estimates Across Regimes

Model	Period	Interest Rate Coefficient	p-value
ECB	Apr 2000 – Jul 2007	0.084	0.020
ECB	Aug 2007 – Feb 2010	1.091	< 0.001
ECB	Mar 2010 – Nov 2016	2.146	< 0.001
ECB	Dec 2016 – May 2020	18.974	< 0.001
ECB	Jun 2020 – Dec 2022	1.470	< 0.001
ECB	Jan 2023 – Feb 2026	-0.827	< 0.001
Fed	Feb 1970 – Nov 1980	0.988	< 0.001
Fed	Dec 1980 – Feb 1987	0.762	< 0.001
Fed	Mar 1987 – Dec 1999	0.577	< 0.001
Fed	Jan 2000 – Oct 2009	-0.185	0.013
Fed	Nov 2009 – Jun 2020	0.627	< 0.001
Fed	Jul 2020 – Feb 2026	-0.931	< 0.001

Notes: The table reports the estimated coefficient on the fitted interest rate from the second-stage equation within each regime identified by the Bai-Perron procedure.

4.3 Result for Instrument Diagnostic Tests

Since the leadership regime model and the structural break analysis are estimated using TSLS, the validity of the instruments is evaluated using diagnostic tests. Instrument relevance and strength are assessed using weak instrument tests reported by the TSLS estimation procedure. For both the ECB and Federal Reserve models, the weak-instrument tests strongly reject the null hypothesis of weak instruments. This indicates that the lagged variables used as instruments are strongly correlated with the endogenous regressors and satisfy the relevance condition. The exogeneity of the instruments is evaluated using the Sargan test of overidentifying restrictions. However, for both models, the Sargan test rejects the null hypothesis that all instruments are exogenous. Therefore, while the instruments appear to be relevant and sufficiently strong, the exogeneity assumption may not be fully

satisfied. The TSLS results should therefore be interpreted with some caution.

Table 14: Instrument Validity Tests

	ECB	Federal Reserve
Weak Instrument Test p-value	< 0.001	< 0.001
Sargan Test Statistic	280.845	591.335
Sargan Test p-value	< 0.001	< 0.001

Notes: The weak-instrument test evaluates instrument relevance and strength. The Sargan test evaluates the exogeneity of the overidentifying restrictions.

5 Discussion

This chapter discusses the main findings presented in the previous results chapter. It begins by focusing on the estimated coefficients on the lagged interest rate and considering possible explanations for their positive signs. Then, it discusses whether the estimates after HAC correction, joint Wald tests and structural break analysis provide evidence that the effect of monetary policy on inflation varies across leadership periods. The structural break results are also compared with leadership shifts and major economic events to figure out the possible reasons behind the estimated break dates. In addition, the chapter compares the ECB and the Federal Reserve to identify similarities and differences in their results and to assess whether the evidence for time-varying effects differs across the two central banks. Finally, it discusses the limitations of the study and gives advice for future research. Together, these discussions help provide a broader interpretation of the empirical findings and their implications for understanding the relationship between monetary policy and inflation.

5.1 Interpreting the Lagged Interest Rate Coefficients

An interesting result is that the estimated coefficient on the lagged interest rate is positive in both the ECB and Federal Reserve models. From a standard monetary policy perspective, higher interest rates are generally expected to reduce inflation by lowering

aggregate demand (Mishkin, 2023). Hence, an inverse relationship between interest rate and inflation is usually expected. The positive coefficients found in this study should not necessarily be interpreted as higher interest rates cause higher inflation. In practice, central banks often raise interest rates when inflation is already increasing. As a result, interest rates and inflation may move in the same direction over certain periods even if the ultimate target of the monetary policy is to reduce inflation. Thus, the estimated coefficients may reflect the timing and way central banks respond to inflation rather than the causal effect of monetary policy itself.

This finding also gives support for treating the lagged interest rate as endogenous. If monetary policy systematically responds to inflation developments, interest rates and inflation are jointly determined rather than independent of each other. This is one reason why the analysis relies on TSLS estimation rather than a simple OLS regression.

5.2 Evidence for Time-Varying Effects

The results from the TSLS models with HAC standard errors support the existence of time-varying effects in the relationship between interest rates and inflation. For the ECB, only the Draghi interaction term remains statistically significant after HAC correction, while the Trichet and Lagarde interactions are not significant. In addition, the joint Wald test does not reject the null hypothesis that all interaction coefficients are jointly equal to zero. Therefore, the evidence for time-varying effects across ECB leadership periods is relatively limited. For the Federal Reserve, several interaction terms remain significant after HAC correction, including those associated with Greenspan and Bernanke. These results suggest that the relationship between interest rates and inflation is not constant across different leadership periods. At the same time, the joint Wald tests reject the null hypothesis that all interaction effects are jointly equal to zero. Although some interaction terms lose significance after HAC correction, the interaction effects remain jointly significant when considered together. Therefore, the evidence for time-varying monetary policy effects is stronger for the Federal Reserve than for the ECB.

The structural break analysis contributes to a similar conclusion. The Bai-Perron algorithm identifies 13 breaks for the ECB and 8 breaks for the Federal Reserve, and the

subsequent Bai-Perron tests reject the null hypothesis of no structural breaks. In addition, the estimated interest-rate coefficients reported in Table 13 vary substantially across the identified regimes. For the ECB, the coefficient increases from 0.084 before the first break to 18.974 during the 2016-2020 period before becoming negative in the final regime. For the Federal Reserve, the coefficient changes from positive values before 2000 to negative values in some later regimes. These results provide further evidence that the effect of monetary policy on inflation is not constant over time.

5.3 Interpreting the Break Dates

Using the 5% significance level and the results for individual Bai-Perron tests reported in Table 12, only a subset of the ECB breaks remains significant. These breaks are in October 2005, August 2008, December 2013, May 2020, August 2022 and October 2022. In contrast, all estimated breaks for the Federal Reserve remain statistically significant when tested individually.

For the ECB, the estimated break dates do not closely match leadership transitions. While several breaks can be connected to economic events, there is no clear evidence that explains the break identified in October 2005. In contrast, the breaks in 2008 and 2013 occur during the Global Financial Crisis and the European debt crisis. Previous studies have argued that these events changed financial conditions and the relationship between monetary policy and inflation in the euro area (Cour-Thimann & Winkler, 2012). The break identified in May 2020 coincides with the COVID-19 pandemic, while the break in August and October 2022 occur during the Russia-Ukraine war periods. Both events were associated with sharp increases in inflation and changes in monetary policy (IMF, 2020; IMF, 2023). Therefore, these estimated breaks may not be so closely connected to leadership transitions.

For the Federal Reserve, the breaks are identified in October 1980, November 1980, February 1987, June 1987, December 1999, October 2009, March 2020 and June 2020. Compared with the ECB, some of these breaks are more closely match with changes in Federal Reserve leadership. The break dates in 1980 appear shortly after Volcker took office and coincide with the beginning of the Volcker disinflation period, during which the

Federal Reserve adopted a sustained restrictive monetary policy and raised the federal funds rate to historically high levels to reduce inflation (Goodfriend & King, 2005). Similarly, the breaks in February and June 1987 are close to the appointment of Greenspan as chairman. However, since both breaks occur before Greenspan took office officially in August 1987, there is limited evidence that they can be exactly explained by the change in leadership. However, not all breaks can be connected to the leadership shifts. The break in October 2009 occurs during the Global Financial Crisis. During this period, the Federal Reserve implemented unusually easy monetary policy, introduced new non-Bagehot liquidity facilities and adopted non-conventional monetary policy by large-scale asset purchases to lower interest rates (Mishkin & White, 2014). Similar to the ECB, break dates in 2020 coincides with the COVID-19 pandemic as well. Only the break in December 1999 occurs during Greenspan's tenure and does not correspond to any leadership changes or monetary policy changes, so there is almost no evidence linking this break to a specific historical activity.

These suggest that leadership changes are not the only origin of time variation. Several estimated breaks coincide more closely with big economic events than with changes in central bank leadership. Therefore, the relationship between interest rates and inflation may be influenced by both leadership regimes and broader macroeconomic developments.

5.4 Comparison between ECB and Federal Reserve

A notable difference can be observed when comparing the estimated leadership effects across the two central banks. For the ECB, only the Draghi interaction term remains statistically significant and positive after accounting for endogeneity and HAC standard errors, implying a stronger positive relationship between interest rates and inflation during the Draghi period. In contrast, several Federal Reserve leadership interactions are negative, although not all of them are statistically significant. These results suggest that the pattern of time variation differs across the two central banks. One possible explanation is that the ECB and the Federal Reserve operated under different macroeconomic conditions during their respective leadership periods. For example, much of the Draghi period was characterized by persistently low inflation in the euro area, whereas several Federal Reserve leadership periods were associated with monetary tightening episodes aimed at containing

inflationary pressures.

Another difference is that the Federal Reserve provides stronger evidence of time variation. For the ECB, only the Draghi interaction term is statistically significant, while the Trichet and Lagarde coefficients are not. In contrast, several Federal Reserve leadership periods show significant differences from the baseline regime. The joint Wald tests provide similar evidence. For the ECB, the null hypothesis that all leadership interaction terms are jointly equal to zero cannot be rejected ($\chi^2 = 5.521$, $p = 0.137$). For the Federal Reserve, however, the null hypothesis is strongly rejected ($\chi^2 = 53.141$, $p < 0.001$). These findings suggest that leadership periods are more strongly associated with differences in the estimated monetary policy effect in the Federal Reserve than in the ECB. One possible explanation is the difference in sample length. The Federal Reserve sample covers more than five decades, while the ECB sample begins only in 2000. As a result, the Federal Reserve data include a wider range of economic conditions, such as the high-inflation period of the 1970s, the Volcker disinflation, the Great Moderation, the Global Financial Crisis and the COVID-19 pandemic. These events were associated with substantial changes in inflation dynamics and monetary policy behavior. In contrast, the ECB sample covers a much shorter period and contains fewer leadership transitions. This may make it more difficult to detect systematic differences across leadership periods. A longer sample also provides more observations and greater variation in the data, which may contribute to the stronger statistical evidence found for the Federal Reserve. Therefore, part of the difference between the ECB and Federal Reserve results may reflect differences in the available sample period rather than differences in leadership effects alone. Another possible explanation is the institutional structure of the two central banks. Monetary policy decisions in the euro area are made collectively within the ECB Governing Council, whereas the Federal Reserve chairman is often more closely associated with the direction of monetary policy. As a result, changes in ECB leadership may not translate into changes in monetary policy behavior to the same extent as changes in Federal Reserve leadership.

5.5 Limitations and Suggestions

This study has several limitations. First of all, although the TSLS estimates are based on lagged macroeconomic variables as instruments, the Sargan tests reject the null hypothesis of instrument exogeneity for both the ECB and Federal Reserve models. While the instruments are highly relevant and sufficiently strong, the validity of the exogeneity assumption may be questioned. Secondly, the analysis focuses on interest rates, unemployment and leadership periods, while other factors that may affect inflation are not included in the model. Inflation is influenced by a wide range of macroeconomic developments, including supply shocks, energy prices and fiscal policy. Therefore, the estimated coefficients should not be interpreted as capturing all determinants of inflation.

Future research could explore alternative instrument sets and identification strategies to address potential concerns regarding instrument exogeneity. In addition, future studies could include a broader set of macroeconomic variables, such as energy prices if the data is available, to better capture the factors affecting inflation. It may also be useful to investigate whether the observed time variation is driven by leadership changes, institutional developments or major macroeconomic events.

6 Conclusion

This paper examined whether the effect of monetary policy on inflation varies over time in the euro area and the United States. Using leadership-based interaction models and structural break analysis, the results provide evidence that the relationship between interest rates and inflation is not constant over time.

For the leadership regime model, the Federal Reserve provides stronger evidence of time-varying effects than the ECB. The joint Wald test rejects the null hypothesis of no leadership effects for the Federal Reserve but not for the ECB, suggesting that the evidence for time-varying effects across ECB leadership periods is relatively weak. The structural break analysis also identifies many significant breaks for both central banks. In addition, the estimated coefficients vary across the regimes identified by the Bai-Perron procedure, indicating that the effect of interest rates on inflation changes over time. Overall, the

findings suggest that monetary policy effects are time-varying, although the evidence is considerably stronger for the Federal Reserve than for the ECB. The results also indicate that leadership changes alone cannot fully explain the observed variation. While some of the estimated breaks occur close to leadership transitions, others coincide more closely with economic events and periods of changing macroeconomic conditions. These findings suggest that the relationship between interest rates and inflation is influenced by multiple factors and may not remain stable over long periods. These findings suggest that the relationship between interest rates and inflation changes over time and cannot always be exactly represented by a constant coefficient. Future research may therefore consider using time-varying frameworks that allow the effect of monetary policy to vary across different periods.

7 References

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8 Appendix

Table 15: BIC Model Selection for the First-Stage Interest Rate Equation

ECB		Federal Reserve	
Breaks (m)	BIC	Breaks (m)	BIC
0	-584.442	0	943.9
1	-583.860	1	862.6
2	-574.167	2	**821.5**
3	**-596.500**	3	862.9
4	-585.300	4	910.9
5	-566.756	5	960.1
6	-531.860	6	1009.4
7	-486.137	7	1059.4
8	-440.271	8	1114.0
9	-293.416	9	1184.9

Notes: The table reports BIC values for alternative numbers of structural breaks estimated using the Bai-Perron procedure in the first-stage interest rate equation. The preferred specification is the one with the lowest BIC value, highlighted in bold. The ECB model selects three breaks, while the Federal Reserve model selects two breaks.

Table 16: BIC Model Selection for the First-Stage Unemployment Equation

ECB		Federal Reserve	
Breaks (m)	BIC	Breaks (m)	BIC
0	-305.787	0	1347.2
1	-299.984	1	**1037.4**
2	-320.855	2	1072.2
3	-363.993	3	1106.7
4	-373.950	4	1145.0
5	-384.927	5	1184.3
6	**-385.886**	6	1232.3
7	-351.479	7	1279.2
8	-309.880	8	1332.2
9	-130.516	9	1401.5

Notes: The table reports BIC values for alternative numbers of structural breaks estimated using the Bai-Perron procedure in the first-stage unemployment equation. The preferred specification is the one with the lowest BIC value, highlighted in bold. The ECB model selects six breaks, while the Federal Reserve model selects one break.

Table 17: BIC Model Selection for the Second-Stage Structural Break Analysis

ECB		Federal Reserve	
Breaks (m)	BIC	Breaks (m)	BIC
0	1188.74	0	2843.0
1	916.77	1	2375.8
2	766.80	2	2030.2
3	682.47	3	1881.3
4	636.22	4	1828.5
5	**603.02**	5	**1725.1**
6	613.36	6	1730.3
7	626.43	7	1745.9
8	671.34	8	1833.5
9	764.11	9	1941.5

Notes: The table reports BIC values for alternative numbers of structural breaks estimated using the Bai-Perron procedure in the second-stage equation. The preferred specification is the one with the lowest BIC value, highlighted in bold. Both the ECB and Federal Reserve models select five breaks.